Neyman-Pearson Classification Problem

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Type I & II Error

Let $\mathcal{D}_n := \{(\boldsymbol{x}_1, y_1), \cdots, (\boldsymbol{x}_n, y_n)\}$ be the training set, $(\boldsymbol{x}_i, y_i)$ are jointly distributed random variables from the probability space $\mathcal{X} \times \mathcal{Y}$. We consider a binary classification problem where $y_i \in \{0, 1\}$. A **classifier** is a Borel-measurable function $f : \mathcal{X} \to \mathcal{Y}$. Our goal is to find the "best" classifier based on the data we have and make predictions on other data.

- Misclassification occurs when $f(\boldsymbol{x}_i) \neq y_i$, it can be summarized into two types of misclassification:
  - **Type I Error:** $\mathbb{P}(f(\boldsymbol{x}_i) = 1 | y_i = 0)$, namely, the conditional probability that the predicted label is 1 given that the true label is 0.
  - **Type II Error:** $\mathbb{P}(f(\boldsymbol{x}_i) = 0 | y_i = 1)$, namely, the conditional probability that the predicted label is 0 given that the true label is 1.
- We use $R_0(f), R_1(f)$ to denote the type I,II error made by classifier f, respectively.

Minimizing the Risk

In order to find the "best" classifier, the classicial binary classification aims to minimize the weighted sum of type I,II error:

$$f^* := \arg\min_f \mathbb{P}(y=0)R_0(f) + \mathbb{P}(y=1)R_1(f)$$

where $\mathbb{P}(y=0), \mathbb{P}(y=1)$ are some priori probabilities of the class $0, 1$.

However, this minimization is not always optimal.

Minimizing the Risk

- Consider an application in cancer diagnosis. If we let $y = 0$ be the class of healthy people, and $y = 1$ be the class of diseased people. Then the concequences of type I & II error may be different, i.e misclassify a diseased person as healthy may be more serious than misclassify a healthy person as diseased!
- In Neyman-Pearson classification, we will manually set up a threshold α, such that we aim to minimize $R_1(f)$ under the constrain that $R_0(f) \leq \alpha$ or vise versa. This is the **NP Oracle Classifier**:

$$f^* = \text{argmin}_{f, R_0(f) \leq \alpha} R_1(f)$$

or

$$f^* = \text{argmin}_{f, R_1(f) \leq \alpha} R_0(f)$$

Neyman-Pearson Lemma

- The NP Oracle Classifier f^* can be obtained by Neyman-Pearson lemma:

$$f^*(\boldsymbol{x}) := \begin{cases} 1 & \text{if } \frac{f_1(\boldsymbol{x})}{f_0(\boldsymbol{x})} > \eta \\ 0 & \text{otherwise} \end{cases}$$

where $f_0(\boldsymbol{x}), f_1(\boldsymbol{x})$ are the density from two classes and η satisfies

$$\mathbb{P}\left(\frac{f_1(\boldsymbol{x})}{f_0(\boldsymbol{x})} > \eta \middle| y = 0\right) = \alpha$$

- So if the density of $f_0(\boldsymbol{x}), f_1(\boldsymbol{x})$ are given, then NP-lemma will give us the optimal classifier for a given significance level α.
- We are more interested in the case when their density is not known, i.e we only know $f_j, j \in \{0, 1\}$ by a training set $\mathcal{D}_n$.
- Two classical techniques are **Empirical Risk Minimization Approach** and **Plug-in Approach**.

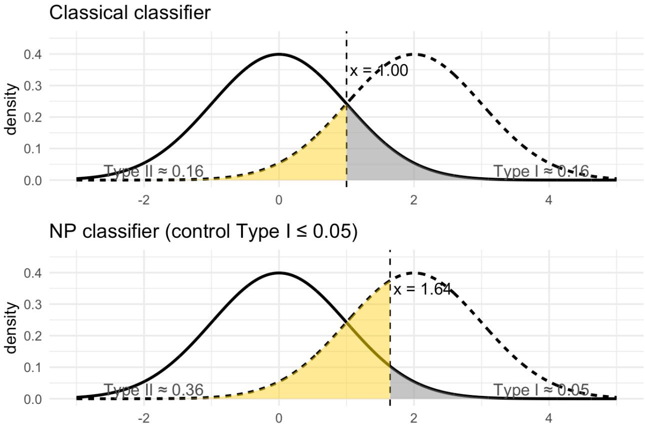

Figure: A simulation: The true distributions of $\boldsymbol{x}$ under two classes are $\mathcal{N}(0,1)$ (with $y=0$), $\mathcal{N}(2,1)$ (with $y=1$). If user prefers a type I error $R_0(f) \leq 0.05$, a classical classifier $\mathbf{1}\{x \geq 1\}$ would have $R_0(f) = 0.16$, but the NP classifier $\mathbf{1}\{x \geq 1.65\}$ will satisfy the type I error constraint while minimize type II error.

Empirical Risk Minimization

We first study the **Empirical Risk Minimization (ERM)** approach. We define

$$n_j = \sum_{i \in [n]} \mathbf{1}\{y_i = j\} \quad j \in \{0, 1\}$$

be the number of samples from class j. Then we define the **empirical error** (or false alarm, miss probability) of the classifier f to be

$$\widehat{R}_j(f) := \frac{1}{n_j} \sum_{i \in [n]: y_i = j} \mathbf{1}\{f(\boldsymbol{x}_i) \neq j\}$$

corresponding to $j = 0, 1$.

Empirical Risk Minimization (Work by Cannon et al.)

Cannon et al. (2002) showed that, the ERM type classifier guarantee the probabilistic bounds for fixed tolerance levels $\epsilon_0, \epsilon_1 > 0$ as follows: Set

$$\widehat{f}_n := \text{argmin}_{f \in \mathcal{H}, \widehat{R}_0(f) \leq \alpha + \epsilon_0/2} \widehat{R}_1(f)$$

where $\mathcal{H}$ is the set of classifiers with finite Vapnik-Chervonenkis (VC) dimension V. Then under *i.i.d sampling* in which $n_), n_1$ are unknown until the training sample is observed, if $n \geq \frac{10\sqrt{5}}{\pi_j^2 \epsilon_j^2}$, we have

$$\begin{aligned}\mathbb{P}\left[\left\{R_0(\widehat{f}_n) - \alpha > \epsilon_0\right\} \cup \left\{R_1(\widehat{f}_n) - R_1(f^*) > \epsilon_1\right\}\right] \\ \leq 10(2n)^V \left[\exp\left(-\frac{n\pi_0^2\epsilon_0^2}{640\sqrt{5}}\right) + \exp\left(-\frac{n\pi_1^2\epsilon_1^2}{640\sqrt{5}}\right)\right].\end{aligned}$$

Empirical Risk Minimization (Work by Cannon et al.)

Another prbabilistic bound can be achieved under *retrospective sampling* where n_0, n_1 are known before the sample is observed, is given by

$$\mathbb{P}\left[\left\{R_0(\widehat{f_n}) - \alpha > \epsilon_0\right\} \cup \left\{R_1(\widehat{f_n}) - R_1(f^*) > \epsilon_1\right\}\right] \\ \leq 8n_0^V e^{-n_0\epsilon_0^2/128} + 8n_1^V e^{-n_1\epsilon_1^2/128}.$$

Empirical Risk Minimization (Work by Scott and Nowak)

Scott and Nowak (2005) pointed out that the above bound under *i.i.d* sampling proposed by Cannon et al. is substantially larger then that for retrospective sampling, and does not hold for small n. They proposed an alternative way to derive the probabilistic bounds which apply to both sampling schemes for all values of n where the tolerance levels ϵ_0, ϵ_1 are variable.

Empirical Risk Minimization (Work by Scott and Nowak)

For any $\delta_0, \delta_1 > 0, n \in \mathbb{N}$,

- For a VC class $\mathcal{H}$ with VC dimension V, let

$$\epsilon_j = \epsilon_j(n_j, \delta_j, \mathcal{H}) = \sqrt{128 \cdot \frac{V \log n_j + \log(8/\delta_j)}{n_j}}, \quad j = 0, 1$$

- For a finite class $\mathcal{H}$, let

$$\epsilon_j = \epsilon_j(n_j, \delta_j, \mathcal{H}) = \sqrt{2 \cdot \frac{\log |\mathcal{H}| + \log(2/\delta_j)}{n_j}} \quad j = 0, 1$$

then the classifier $\widehat{f}_n$ satisfies

$$\mathbb{P}\left[\left\{R_0(\widehat{f}_n) - \alpha > \epsilon_0\right\} \cup \left\{R_1(\widehat{f}_n) - R_1(f^*) > \epsilon_1\right\}\right] \leq \delta_0 + \delta_1.$$